A REPRESENTABLE MORPHISM TO AN ALGEBRAIC STACK HAS ALGEBRAIC SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span. The transfer inputs are not carded and are declared external.

Allowed standard prerequisites (editorial).

- Base change by a space-representable category. lemma-base-change-by-space-representable-by-space (carded L26677). U = V x_Y X is representable by an algebraic space.
- Base-change stability. lemma-base-change-representable-by-spaces (same chapter, NOT carded, declared external). U -> X is representable by algebraic spaces.
- Base-change property transfer. lemma-base-change-representable-transformations-property (same chapter, NOT carded, declared external). U -> X stays smooth and surjective.
- Smooth surjective implies algebraic. lemma-smooth-surjective-morphism-implies-algebraic (carded L26688). Final step.

Source reference. Original definition 026O: sources/stacks-algebraic/pinned-original-source.tex, source lines 1506–1506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0300: sources/stacks-algebraic/pinned-original-source.tex, source lines 900–900. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0302: sources/stacks-algebraic/pinned-original-source.tex, source lines 876–876. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045C: sources/stacks-algebraic/pinned-original-source.tex, source lines 1180–1180. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05UL: sources/stacks-algebraic/pinned-original-source.tex, source lines 2082–2082. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X} \to \mathcal{Y}$ be a morphism of stacks in groupoids over $(Sch/S)_{fppf}$. Assume that*

(1) *$\mathcal{X} \to \mathcal{Y}$ is representable by algebraic spaces, and*
(2) *$\mathcal{Y}$ is an algebraic stack over S.*

Then $\mathcal{X}$ is an algebraic stack over S.

Proof. Let $\mathcal{V} \to \mathcal{Y}$ be a surjective smooth 1-morphism from a representable stack in groupoids to $\mathcal{Y}$. This exists by Definition 026O. Then the 2-fibre product $\mathcal{U} = \mathcal{V} \times_{\mathcal{Y}} \mathcal{X}$ is representable by an algebraic space by Lemma 0300. The 1-morphism $\mathcal{U} \to \mathcal{X}$ is representable by algebraic spaces, smooth, and surjective, see Lemmas 0302 and 045C. By Lemma 05UL we conclude that $\mathcal{X}$ is an algebraic stack. □